

Currency Converter App Documentation

Project Overview

This Currency Converter application is developed using Kotlin, Jetpack Compose, Retrofit API Integration, and State Management. The application allows users to convert currencies using real-time exchange rates obtained from an online currency API.

Project Scope

The application provides currency conversion functionality with real-time exchange rates. Users can enter an amount, select a base currency, view converted values, and access a complete list of exchange rates for different currencies.

Technologies Used

- Kotlin
- Jetpack Compose
- Retrofit
- MVVM Architecture
- State Management (StateFlow / Compose State)
- Currency Exchange API

Features

1. Currency Conversion
2. Real-Time Exchange Rates
3. View All Exchange Rates
4. Base Currency Selection
5. API Integration
6. Dynamic UI Updates

Core Functionalities

Convert: Convert an entered amount from the selected base currency into multiple currencies.

Read: Display current exchange rates from the API.

View Rates: Show detailed exchange rate information on a separate screen.

API Integration: Fetch real-time currency data from the server.

State Management

The UI automatically updates whenever new exchange rate data is received from the API. State management ensures synchronization between the ViewModel, API responses, and the user interface.

Application Workflow

1. User enters an amount.
2. User selects a base currency (USD, INR, EUR, etc.).
3. Application sends a request to the Currency Exchange API.
4. Exchange rates are received from the server.
5. Converted currency values are calculated.
6. Results are displayed on the Home Screen.
7. User can navigate to the Exchange Rates Screen to view complete exchange rate details.

Screens

Home Screen

- Enter Amount
- Select Base Currency
- Convert Currency
- Display Converted Results

Exchange Rates Screen

- Display Complete Exchange Rates
- Base Currency Information

- Exchange Rate Details

Architecture

UI Layer (Jetpack Compose)

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ViewModel Layer

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Repository Layer

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Retrofit API Service

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Currency Exchange API

Advantages

- Simple and user-friendly interface
- Real-time currency conversion
- Fast API-based data retrieval
- Clean MVVM Architecture
- Easy to maintain and extend

Future Enhancements

- Currency Dropdown Menus
- Conversion History
- Favorites
- Offline Support using Room Database
- Currency Search

- Dark Mode Support